

Longest Increasing Subsequence

3 1 4 5 9 2 6 5 3 5 8 9 7

1 4 5 6 7

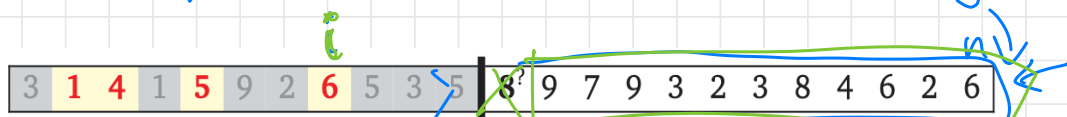
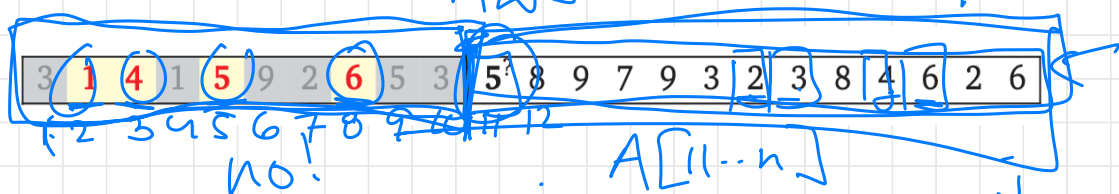
subsequence

9 2 6 5

substring
contiguous subsequence

choice: do we include next element?

$A[i] > A[j]$



to explore choosing and not choosing next elt:

$A[i]$ last element chosen

$A[j..n]$ suffix remaining

impl step: define recursive subproblem

$LIS(i, j)$ = length of the longest ~~incr.~~ subsequence in $A[j..n]$ using only numbers ~~greater~~ than $A[i]$.

$LIS(i, j)$

$$LIS(i, j) = \begin{cases} 0 \\ \max \begin{cases} LIS(i, j+1) \\ L(j, j+1) + 1 \\ LIS(i, j+1) \end{cases} \end{cases}$$

if $j > n$
if $A[i] > A[j]$
otherwise

2 choices: incl j
don't incl j

$$LIS(8, 11) \geq 4$$

$LIS(i, j)$:
LDS

if $j > n$:

return 0

if $A[i] > A[j]$:

$LIS(i, j+1)$

else:

return $\max(LIS(j, j+1) + 1, LIS(i, j+1))$

incl j
don't incl j

$LAS(i, j, k)$

last selected start of suffix boolean: use
need smaller need larger

$LDS'(i, j)$
 $LIS'(i, j)$

can't incl j

with table:
update ①, ②, ③ for longest decreasing subseq
if time: longest alternating subseq

